

Reward Model Training

Bradley-Terry pairwise preferences $\rightarrow$ scalar reward $r(x,y)$: the proxy for human judgment

10 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

AI-RL-IN-TH-000002

ID

Reward Model — What It Is and How It's Built

Definition

The reward model $r(x,y)$ is a neural network that takes a (prompt, response) pair as input and outputs a scalar score representing how much a human would prefer this response. It is trained on pairwise human preference data.

- Architecture: the SFT model with the LM head (vocabulary logits) replaced by a scalar regression head (linear layer $\rightarrow$ single scalar)
- Training data: for each prompt x , N responses $\{y_1 \dots y_n\}$ are generated; human annotators rank them; $N(N-1)/2$ pairwise comparisons are extracted
- Bradley-Terry model: $P(y_w > y_l \mid x) = \sigma(r(x, y_w) - r(x, y_l))$ — the probability that response y_w is preferred over y_l
- Loss: binary cross-entropy $L = -E[\log \sigma(r(x, y_w) - r(x, y_l))]$

Bradley-Terry Model — Mathematical Derivation

Given: prompt x , winning response y_w , losing response y_l

Preference probability: $P(y_w \succ y_l \mid x) = \sigma(r(x, y_w) - r(x, y_l))$

where $\sigma(z) = 1 / (1 + e^{-z})$ is the sigmoid function

Reward model training loss (binary cross-entropy on pairs):

$$L_{RM} = -E_{\{(x, y_w, y_l) \sim D\}} [\log \sigma(r_{\theta}(x, y_w) - r_{\theta}(x, y_l))]$$

Intuition:

Maximise the score gap ($r_w - r_l$) for human-preferred pairs. The RM learns to assign higher scores to responses humans prefer.

From N responses: $n(n-1)/2$ pairs extracted $\rightarrow$ exponential preference signal

4 responses per prompt $\rightarrow 4 \times 3 / 2 = 6$ comparison pairs per prompt

Reward Model Failure Modes — What Can Go Wrong

Verbosity Bias

Annotators tend to prefer longer responses regardless of quality. RM learns to score verbose outputs higher — leading to padding in the final model.

Position Bias

Annotators rate the first response higher when shown two options side-by-side. Counteract: randomise response order during annotation.

Sycophancy Bias

Annotators prefer responses that agree with their stated beliefs. RM learns to produce agreeable but inaccurate outputs.

Annotator Disagreement

Inter-annotator agreement typically 60–75%. Low agreement → noisy labels → noisy RM. Mitigate with annotation guidelines and calibration sessions.

KEY REFERENCES

- Ouyang, L. et al. (2022). InstructGPT. NeurIPS 2022.
- Bradley, R.A. & Terry, M.E. (1952). Rank Analysis of Incomplete Block Designs. Biometrika.
- Lambert, N. (2024). RLHF Book. rlhfbook.com

 **Next: Concept 21: Policy Optimisation (AI-RL-IN-TH-000003)**